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A Data Scientist's Guide to Communicating Results



Cecelia Shao

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So your model is finally done running, you've tweaked and optimized all of the hyperparameters you could to obtain the best results, and you're ready to present your findings. Now what?

One of the most important skills for data scientists to have is being able to clearly communicate results so different stakeholders can understand. Since data projects are collaborative across functions and data science results are often incorporated into a larger final project, the true impact of a data scientists' work depends on how well others can understand their insights to take further action.

Here at [Comet.ml](https://comet.ml), we strive to make make this process of communicating both results and the steps leading up to those results easier 👍

cometpublic / home-credit / Experiment (ade76bf82bf1488e9e9cf40d7168f9fa) [Create Pull-Request](#)

Status	File name	Commit SHA	Server start time	Duration
✓	Notebook	N/A	6/29 05:03 PM	145:46:39

Charts
<> Code
Hyper parameters
Metrics
Graph definition
Output
Installed packages
Notes
Graphics
Other

File Edit Insert View Format Table Tools saved a few seconds ago

Paragraph

Dataset from Kaggle (home credit): <https://www.kaggle.com/c/home-credit-default-risk>

Uploaded multiple figures (see graphics) from EDA

Reduce feature space --> 40 to 5-10 features using dimensionality reduction techniques (features like # of certain types of rooms) + make more plots

Model options to try on split data:

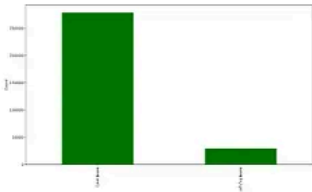
1) simple NN as a feature extractor

2) XGBoost/LGBM

****Remember:** huge ensembles usually win @ Kaggle competitions

Exploratory Data Analysis:

The contract data looks very skewed. Would recommend looking at this article on [dealing with skewed data](#)



Take detailed notes in [Comet.ml](#) for your teammates and managers — make sharing and communicating results easy!

In this post, we'll explore:

1. Who might be in your audience
2. How to effectively structure your presentation and results based on your audience
3. Common errors with communication to watch out for
4. How Comet.ml helps with communication

Who is your audience?

Throughout every project, you'll encounter people with varying levels of technical expertise, buy-in, and business goals. When you present to these different stakeholders, make sure you keep an eye on how your work ties into their role and decisions.

Here are three common audiences:

- **Your team manager:** probably the first line of review for any work you do or show to other stakeholders. Your manager may or may not be technical, but they certainly will be communicating with other teams/stakeholders.
- **Line-of-business (LOB) stakeholders:** this could be a product manager, business analyst, or a VP of customer support. Data science is amazing because it enables cross-functional work — just remain aware of how your insights or recommendations influence other teams' workflows.
- **Data engineers/engineering team:** don't forget the team that's working to deploy champion models! Just because these are more technical stakeholders doesn't necessarily mean they should not have any business context in the information — often times

Making a targeted presentation

Once you understand your audience, you can begin tailoring an effective and targeted presentation.

With non-technical stakeholders, you should avoid highly technical terms (e.g. your hyperparameters for your TensorFlow model) and instead, try to frame the machine learning problem into the same terms in which business decisions are made — marginal cost and benefit. However, with your

engineering or devOps team, they will need to know details such as how long the model takes to train and GPU/CPU metrics during training.

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The most important thing to recognize is that this should not be the only time these results are communicated. Frequent communication and feedback will help alleviate pressure on the final presentation, increase buy-in for your work, and help ease business stakeholders into technical details.

Here's a useful starting framework you can use:

1. Your understanding of the business problem
2. How to measure business impact — what business metrics do your model results align to?
3. What data is/was available — if appropriate, reference what data it would be helpful to collect
4. The initial solution hypothesis
5. The solution/model — use examples and visualizations
6. The business impact of the solution and clear action items for stakeholders

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Hungry for more? We also recommend reading these great posts that include tips on communication: (1) [Aspiring Data Scientists? Master these fundamentals](#) from [Peter Gleeson](#) and (2) [The Data Science Process: What a data scientist actually does day-to-day](#) from [Raj Bandyopadhyay](#)

Communicating data science: A guide to presenting your work

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blog.kaggle.com

Common errors to watch out for:

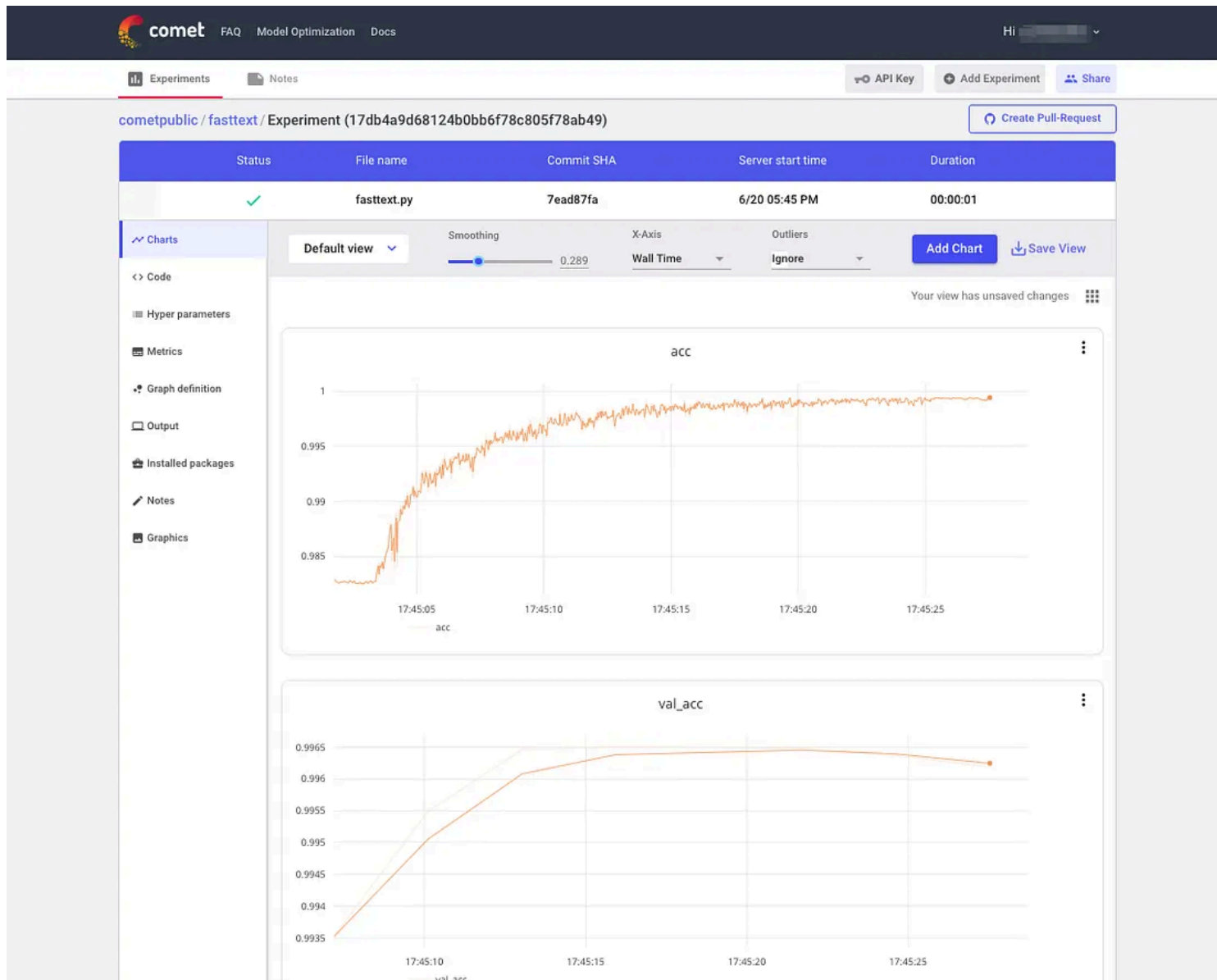
While there are [a number of statistical and technical errors](#) you can make during your analysis, we'll focus on some common communication errors you might run into:

- Omitting/**glossing over any key assumptions** made during the analysis
- Recycling the same presentation for different audiences
- Showing visualizations like charts and tables without re-iterating the main idea
- **Saving all insights *until a final presentation* instead of making the process piecemeal and iterative**
- Saving the findings *until the end of the presentation* — make sure to include an executive summary and recommendations at the beginning of the presentation

- Not having back-up materials of different technical levels — an appendix with supporting details is useful for both the actual presentation itself and context if the presentation is shared
- Not opening up after the presentation (either verbally or via email) for **feedback** — everyone has a different way of absorbing information so if you need to adjust, the easiest way to find out that out is to ask for feedback!

How Comet.ml helps communicate results

At [Comet.ml](https://comet.ml), we help data scientists and machine learning engineers to automatically track their datasets, code, experiments and results creating efficiency, visibility and reproducibility.



Instantly visualize model metrics in Comet.ml — you can export these charts for your presentations!

With Comet.ml, you can visualize and track all your model results — this is especially helpful for long running experiments, since you can track the results *live*. Comet.ml also allows you to:

- Share experiment results via slack or email
- Add rich documentation of your work in context (with images!)

- Visualize and save sampled predictions such as images and figures
- Match each experiment with a dataset hash.

Want to see an example? Follow along as our team competes in Kaggle's Home Credit Default Risk competition in our public project!

Just like you need to iterate with your models to improve their performance, you shouldn't expect to have a perfect presentations skills from the jump. Improving your communication skills requires flexibility for your audiences and practice over time! 🙌

Have your own pro-tips for communicating results that we should feature here? Let us know on twitter @Cometml 🙌

Cecelia Shao is the Product Lead at Comet.ml. 🚀🚀🚀

Before Comet.ml, she was a product manager for IBM Watson's machine learning platform and operated at several startups including Mattermark and Gemini. Communicating results and recommendations from data is her speciality as a user-driven PM.

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Written by Cecelia Shao

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Responses (1)



Write a response

What are your thoughts?



Gary Vartanian

Nov 15, 2023



Solving a really big problem in the space. Thank you



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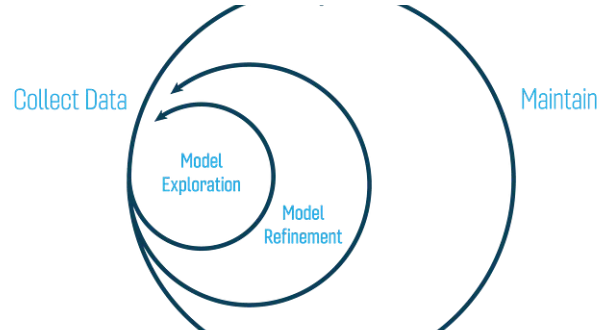


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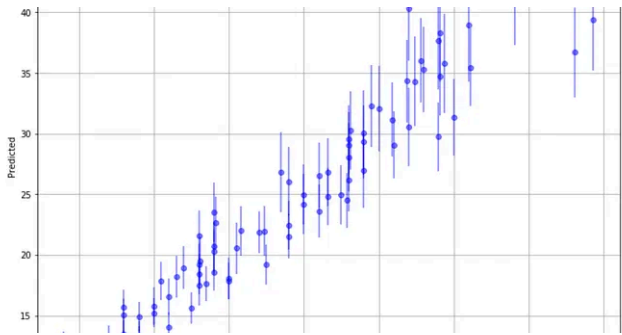
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ABSTRACT

Post-training alignment often reduces LLM diversity, leading to a phenomenon known as *mode collapse*. Unlike prior work that attributes this effect to algorithmic limitations, we identify a fundamental, pervasive data-level driver: *typicality bias* in preference data, whereby annotators systematically favor familiar text as a result of well-established findings in cognitive psychology. We formalize this bias theoretically, verify it on preference datasets empirically, and show that it plays a central role in mode collapse. Motivated by this analysis, we introduce *Verbalized Sampling* (VS), a simple, training-free prompting strategy to circumvent mode collapse. VS prompts the model to verbalize a probability distribution over a set of responses (e.g., “Generate 5 jokes about coffee and their corresponding probabilities”). Comprehensive experiments show that VS significantly improves performance across creative writing (poems, stories, jokes), dialogue simulation, open-ended QA, and synthetic data generation, without sacrificing factual accuracy and safety. For instance, in creative writing, VS increases diversity by 1.6-2.1× over direct prompting. We further observe an emergent trend that more capable models benefit more from VS. In sum, our work provides a new data-centric perspective on mode collapse and a practical inference-time remedy that helps unlock pre-trained generative diversity.

Problem: Typicality Bias Causes Mode Collapse

Typical prompts collapse to different modes:

Solution: Verbalized Sampling (VS) Mitigates Mode Collapse

Different prompts collapse to different modes:



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The Best AI Tools for 2026

If you're going to learn a new AI tool, make sure it's one of these



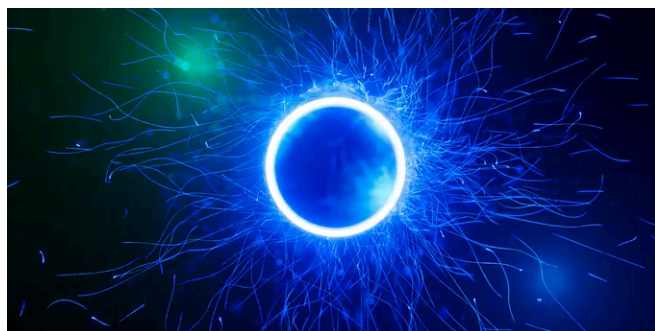
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Will Lockett

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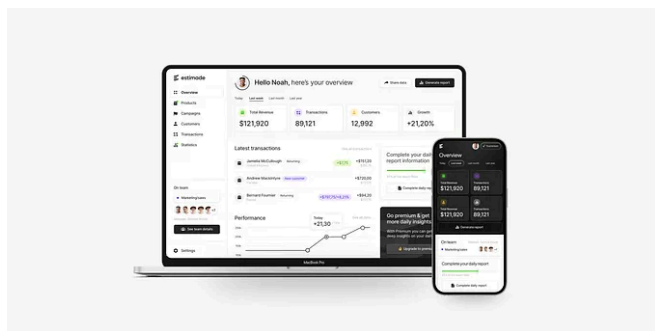
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Michal Malewicz

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